用图表讲故事：叙事可视化策略与技术

Telling a Story with Graphs: How to Apply Narrative Visualization Strategies and Visualization Techniques in Journalistic Storytelling

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用图表讲故事：如何在新闻叙事中运用叙事可视化策略与可视化技术

Telling a Story with Graphs: How to Apply Narrative Visualization Strategies and Visualization Techniques in Journalistic Storytelling

赵鹏

引言

[P001]

英文原文：

Storytelling enables us to arrange our experiences in our memories, convey meaning, and construct and structure our visions of the world we strive to know (Vázquez-Herrero & de Haan, 2022). Journalists rely on storytelling to represent reality and communicate it to the public. The irruption of information technologies has placed journalism in rapid changes and renovated the storytelling formats and models. With the empowerment of digital technologies, journalists have started to be more creative and strategic in presenting their stories, incorporating moving images, sound, videos, interactive data visualization, animations, infographics, multimedia elements, and immersive technology in their storytelling.

中文译文：

讲故事使我们能够在记忆中组织自身经历、传达意义，并建构和组织我们努力认识的世界图景（Vázquez-Herrero & de Haan，2022）。新闻工作者依靠叙事来呈现现实并将其传达给公众。信息技术的迅速兴起使新闻业发生快速变化，也革新了讲故事的形式和模式。借助数字技术，新闻工作者开始以更具创造性和策略性的方式呈现故事，在叙事中融入动态图像、声音、视频、交互式数据可视化、动画、信息图、多媒体元素和沉浸式技术。

[P002]

英文原文：

Of these digital storytelling forms, data journalism, or data storytelling, has gained prominence in both big and small news media organizations. Data journalists heavily rely on large sets of data to write news stories and use data visu-alizations to communicate the final story (Stalph & Heravi, 2023; Weber & Rall, 2012). Data journalists incorporate static or dynamic graphics to tell complex stories, provide insights, reveal trends and patterns, make stories more accessible, and enable audiences to explore datasets (Rodríguez et al., 2015). Nearly 99% of projects nominated by Data Journalism Awards between 2013 and 2016 included data visualizations, which indicates that data visualizations have become an integral part of data-driven stories (Loosen et al., 2020). Storytelling with data visualizations is not simply embedding

bar charts within texts. Without storytelling techniques, data visualizations would not be able to achieve the goal of successful storytelling. Storytelling with data visualization differs from storytelling with text in essential ways. For example, stories in text are typically presented either in a series of linear sequences or in an inverted pyramid structure, which is strictly organized by the author. In contrast, stories in data visualization can be organized in a linear sequence (author-driven) but can also be presented in a non-linear sequence and interactive, inviting audiences to discover the path of the story (reader-driven) (Rodríguez et al., 2015; Segel & Heer, 2010).

中文译文：

在这些数字叙事形式中，数据新闻（data journalism）或数据叙事（data storytelling）在大型和小型新闻媒体机构中都日益受到重视。数据新闻工作者高度依赖大规模数据集来撰写报道，并利用数据可视化呈现最终的故事（Stalph & Heravi，2023；Weber & Rall，2012）。他们将静态或动态图形融入报道，以讲述复杂的故事、提供洞见、揭示趋势与模式，使故事更容易理解，并让受众能够探索数据集（Rodríguez et al.，2015）。2013年至2016年间获得数据新闻奖提名的项目，近99%包含数据可视化，这表明数据可视化已经成为数据驱动叙事不可或缺的一部分（Loosen et al.，2020）。运用数据可视化讲故事，并不只是把柱状图嵌入文字。缺乏叙事技巧，数据可视化就无法实现成功讲故事的目标。数据可视化叙事与文字叙事存在本质区别。例如，文字故事通常以一系列线性顺序或倒金字塔结构呈现，由作者严格组织。相比之下，数据可视化中的故事既可以按线性顺序组织（作者驱动），也可以采用非线性和交互式顺序，邀请受众自行发现故事的展开路径（读者驱动）（Rodríguez et al.，2015；Segel & Heer，2010）。

[P003]

英文原文：

Regarding data visualizations in news stories, journalism scholars focused on data journalism practices (Engebretsen et al., 2018; Heravi & Lorenz, 2020), interactive features (Appelgren, 2018), data visual elements (Stalph & Heravi, 2023), and solutions for historically archiving interactive data visualizations (Heravi et al., 2022). In comparison to studies from the fields of computer science, infographics, and human-computer interaction, studies from journalism scholarship rarely talk about or study the strategies and techniques of incorporating data visualizations in journalistic storytelling. Building on the literature on narrative visualization and visualization rhetoric, this chapter aims to summarize and explain how narrative visualization design strategies and visualization techniques are being used in data visualization to improve journalistic storytelling and discuss how data and digital journalists can benefit by using such techniques to enhance audiences' interpretations of data visualization stories.

中文译文：

在新闻报道中的数据可视化方面，新闻学研究者关注数据新闻实践（Engebretsen et al.，2018；Heravi & Lorenz，2020）、交互功能（Appelgren，2018）、数据视觉元素（Stalph & Heravi，2023），以及交互式数据可视化的历史存档解决方案（Heravi et al.，2022）。与计算机科学、信息图和人机交互领域的研究相比，新闻学研究很少讨论或考察如何在新闻叙事中融入数据可视化的策略和技术。本章以叙事可视化和可视化修辞的文献为基础，旨在总结并解释叙事可视化设计策略及可视化技术如何被用于数据可视化，以改进新闻叙事；同时讨论数据新闻和数字新闻工作者如何利用这些技术，增强受众对数据可视化故事的理解与阐释。

从倒金字塔到创新叙事

[P004]

英文原文：

Advances in technology have pushed the evolution of journalism, bringing along negotiations of journalistic rules, norms, values, practices, and storytelling formats (Song, 2018). Storytelling in

journalism refers to the narrative structure of news (Emde et al., 2016). The affordances of new technologies enabled journalists to pursue innovative modes of storytelling, which shifted from non-participatory and distancing to a more personal and interactive narrative mode (Wahl-Jorgensen & Schmidt, 2019). Although new modes of storytelling challenge traditional journalism norms and practices, it also presents new opportunities for connecting with readers.

中文译文:

技术进步推动了新闻业的演变，同时引发了对新闻规则、规范、价值、实践和叙事形式的协商与调整（Song，2018）。新闻叙事指新闻的叙事结构（Emde et al.，2016）。新技术提供的可供性使新闻工作者能够探索创新的叙事模式，使叙事从非参与式、保持距离的方式，转向更加个人化和互动化的模式（Wahl-Jorgensen & Schmidt，2019）。尽管新叙事模式挑战了传统新闻规范与实践，但也为联系读者带来了新的机会。

[P005]

英文原文:

The inverted pyramid (IP) structure has been a trademark and standardized routine in journalism, for its emphasis on objectivity and constraints on journalists' own values and emotions (Shoemaker & Reese, 1996; Nee & Santana, 2022). The rise of IP is due to the wide adoption of the telegraph at the beginning of the nineteenth century (Canavilhas, 2007). Back in that time, journalists sent news in paragraphs via telegraph, in which information was sent in descending order of importance (Pöttker & Starck, 2003). Such information delivery created the specific journalistic storytelling structure-inverted pyramid, which is contrary to the chronological narrative structure (Norambuena et al., 2020). Structuring information in IP style is important for stories characterized by high news value as the IP structure highlights fact-based reporting, objectivity, and neutrality rather than journalistic interpretation in news writing (Nee & Santana, 2022). Given this advantage, the IP structure is not only popular in print journalism but also adopted by broadcast journalism and, later, online journalism.

中文译文:

倒金字塔（inverted pyramid，IP）结构一直是新闻业的标志性结构和标准化惯例，因为它强调客观性，并对新闻工作者自身的价值判断和情感施加约束（Shoemaker & Reese，1996；Nee & Santana，2022）。倒金字塔的兴起源于19世纪初电报的广泛应用（Canavilhas，2007）。当时，记者通过电报逐段发送新闻，信息按照重要程度递减的顺序传递（Pöttker & Starck，2003）。这种信息传输方式形成了特定的新闻叙事结构——倒金字塔，与按时间顺序展开的叙事结构相反（Norambuena et al.，2020）。对于新闻价值较高的报道而言，按倒金字塔方式组织信息十分重要，因为它强调基于事实的报道、客观性和中立性，而非新闻写作中的记者阐释（Nee & Santana，2022）。鉴于这一优势，倒金字塔不仅流行于报刊新闻，也被广播电视新闻以及后来的网络新闻所采用。

[P006]

英文原文:

Despite the IP's dominance in news writing, the emergence of new technology has prompted scholars and journalists to question and re-examine the efficacy of IP in news delivery (Ytreberg, 2001). Scholars criticized the IP structure for being too strict and formulaic and only representing one end of a story and leaving the other end open (Kosara et al., 2023). Reading IP-structured stories also relies on significant cognitive resources as audiences need to access previously stored knowledge to be able to finish the story (DeAngelo & Yegiyan, 2019). While the IP structured stories require considerable cognitive capacity, available resources are insufficient for audiences to fully understand the entire story (Kleemans et al., 2018). Machill et al. (2007) conducted an experiment to test the impact of two narrative structures—IP

and linear narrative, on audiences' news understanding and engagement. Results showed that linear narrative-structured stories were perceived as more effective and engaging in conveying information.

中文译文：

尽管倒金字塔结构在新闻写作中占据主导地位，新技术的出现仍促使学者和记者质疑并重新审视其传递新闻的效果（Ytreberg，2001）。学者批评该结构过于严格和程式化，只呈现故事的一端，另一端则保持开放（Kosara et al.，2023）。阅读倒金字塔结构的报道还需要大量认知资源，因为受众必须调取已有知识才能读完故事（DeAngelo & Yegiyan，2019）。这类报道需要相当大的认知能力，而受众可用的资源不足以充分理解整个故事（Kleemans et al.，2018）。Machill 等人（2007）通过实验检验了两种叙事结构——倒金字塔和线性叙事——对受众新闻理解及参与程度的影响。结果表明，线性叙事结构被认为在传达信息方面更有效，也更能吸引受众。

[P007]

英文原文：

The introduction of digital media and the Internet brought along new modes of storytelling. News storytelling still features a non-linear narrative form. However, different from the IP structure, storytelling in the age of digital media allows audiences to experience and explore the story in a variety of narratives (Pavlik & Pavlik, 2017). News stories are no longer limited to text or images. Journalists can take advantage of digital media's unique characteristics and integrate a plethora of media formats, including audio, video, data visualizations, graphic animations, virtual/augmented reality, computer algorithms, and games, into news storytelling, helping convey information in an interactive, immersive, and engaging way (Pavlik & Pavlik, 2017; van Krieken, 2018). Some news media developed game formats for news stories that allow audiences play against a computer algorithm to explore and understand the people, events, and contexts in the news event (Pavlik & Pavlik, 2017). Virtual reality and augmented reality are also being experimented with in online news stories to create a more immersive experience.

中文译文：

数字媒体和互联网的引入带来了新的叙事模式。新闻叙事仍具有非线性的形式。然而，与倒金字塔结构不同，数字媒体时代的叙事使受众能够通过多种叙事方式体验和探索故事（Pavlik & Pavlik，2017）。新闻故事不再局限于文字或图片。新闻工作者可以利用数字媒体的独特特性，将音频、视频、数据可视化、图形动画、虚拟现实／增强现实、计算机算法及游戏等多种媒介形式融入新闻叙事，以互动、沉浸和引人入胜的方式传递信息（Pavlik & Pavlik，2017；van Krieken，2018）。一些新闻媒体将新闻故事设计为游戏，让受众与计算机算法对抗，从而探索和理解新闻事件中的人物、事件与背景（Pavlik & Pavlik，2017）。虚拟现实和增强现实也正在网络新闻故事中接受试验，以创造更具沉浸感的体验。

[P008]

英文原文：

Of new emerging journalistic storytelling, data storytelling, which involves using data to uncover insights and tell compelling stories, is becoming increasingly prominent in digital journalism with wide adoption by small and large media outlets. Data storytelling is not simply conveying information in numbers or charts, but telling a story that is comprehensive, memorable, and engaging. Applying data storytelling principles and techniques can improve audiences' recall and engagement of information presented in visualizations (Liem et al., 2020). Therefore, for the rest of this chapter, I will focus on the strategies and techniques that can help journalists improve the effect of data storytelling.

中文译文：

在新兴新闻叙事形式中，数据叙事即利用数据发现洞见并讲述引人入胜的故事，随着大大小小媒体机构的广泛采用，在数字新闻中日益突出。数据叙事不只是用数字或图表传递信息，而是讲述一个全面、令人难忘且

吸引人的故事。运用数据叙事的原则与技巧，可以改善受众对可视化信息的记忆与参与程度（Liem etal., 2020）。因此，本章余下部分将重点讨论能够帮助新闻工作者提高数据叙事效果的策略与技术。

数据叙事与数据可视化

[P009]

英文原文：

Data journalism is a type of journalism where reporters apply the skills and knowledge of data science to journalism (Howard, 2014). The data science skills and knowledge include gathering and validating data as a source, using statistical knowledge to interrogate data, and visualizing data to support the news story. Many journalism practitioners and scholars view data journalism as a way to improve journalism to be more systematic, accurate, and trustworthy (Borges-Rey, 2017; Boyles & Meyer, 2017). Data journalism also blurred the boundary of the professional domain, as journalists, data scientists, designers, and computer scientists are all playing significant parts in data story creation. Data stories, therefore, are multimodal artifacts that integrate together texts, visuals, codings, and design into a coherent whole (Weber et al., 2018).

中文译文：

数据新闻是一种新闻形式，记者在其中将数据科学的技能与知识应用于新闻工作（Howard，2014）。这些技能和知识包括收集和核实作为消息来源的数据、运用统计知识分析数据，以及通过数据可视化支持新闻报道。许多新闻从业者和学者认为，数据新闻能够使新闻报道更加系统、准确和可信（Borges-Rey，2017；Boyles & Meyer，2017）。数据新闻也模糊了专业领域的边界，因为记者、数据科学家、设计师和计算机科学家都在数据故事的创作中发挥重要作用。因此，数据故事是将文本、视觉元素、代码和设计整合为连贯整体的多模态作品（Weber et al.，2018）。

[P010]

英文原文：

Data storytelling is a structured approach by which authors arrange factual pieces in a meaningful order and represent data in a visual format to communicate insights and facilitate understanding (Edmond & Bednarz, 2021; Ojo & Heravi, 2017). The process of creating a data story comprised three major phases: exploring data, making a story, and telling a story. Exploring data refers to activities of exploring and analyzing the raw data that constitutes the content of a story (Lee et al., 2015). The purpose of exploring data is to collect data excerpts that contain insights, variations, or externalization of data. In the story-making process, journalists need to sequence the collected data excerpts into an interesting and compelling storyline. Not only are journalists just ordering data excerpts, they also need to build logical connections, develop flow, and formulate the main message (Lee et al., 2015). Through this phase, journalists should be able to communicate clearly with the story about how the data excerpts are logically related and what all these mean in the big context. Lee et al. (2015) proposed that the final phase includes transforming the abstract content into visual representations, sharing the story, and receiving feedback from audiences.

中文译文：

数据叙事是一种结构化方法，作者将事实材料按有意义的顺序编排，并以视觉形式呈现数据，从而传递洞见并促进理解（Edmond & Bednarz，2021；Ojo & Heravi，2017）。创作数据故事的过程包含三个主要阶段：探索数据、构建故事和讲述故事。探索数据指探索和分析构成故事内容的原始数据（Lee et al.，2015），目的是收集包含洞见、变化或数据外化呈现的数据片段。在构建故事的过程中，记者需要将收集到的数据片段编排成有趣而引人入胜的故事线。他们不仅要排列数据片段，还要建立逻辑联系、形成流畅的叙事，并提炼核心信息（Lee et al.，2015）。经过这一阶段，记者应当能够通过故事清楚说明数据片段之间的逻辑关

系，以及这些片段在更大背景下的意义。Lee 等人（2015）提出，最后一个阶段包括将抽象内容转化为视觉表征、分享故事，以及接收受众反馈。

[P011]

英文原文：

Data visualization is defined as any methodology that can represent data in a visual format that showcases insights and facilitates understanding (Engebretsen et al., 2018; Kirk, 2012). Different from information graphics, data visualization is deeply rooted in quantified facts (Friendly, 2008) and uses geometrical shapes and colors to map out data values and relations (Engebretsen et al., 2018). A well-designed data visualization not only catches audiences' attention but also cuts through the clutter of a complicated story to get right to the point. As advanced technology enabled data visualizations to have an interactive and dynamic interface, the role of data visualization gradually extended to storytelling. Lopezosa et al. (2023) posited that data visualization occupies "a prominent place in storytelling" (p. 315).

中文译文：

数据可视化是指以视觉形式呈现数据、揭示洞见并促进理解的任何方法（Engebretsen et al., 2018; Kirk, 2012）。不同于一般的信息图，数据可视化深深植根于量化事实（Friendly, 2008），并使用几何形状和颜色来映射数据的数值及关系（Engebretsen et al., 2018）。设计良好的数据可视化不仅能够吸引受众注意，还能排除复杂故事中的干扰，直达要点。随着先进技术使数据可视化具备交互式和动态界面，其作用逐渐扩展至讲故事。Lopezosa 等人（2023）认为，数据可视化在叙事中"占据突出地位"（第315页）。

[P012]

英文原文：

Data visualizations serve as a potent tool and provide multiple functions for news storytelling (Engebretsen et al., 2018; Lopezosa et al., 2023). If using data visualizations as a central element in a story, it can emphasize a specific journalistic angle. When claims in a news story need facts to back them up, data visualizations can provide empirical evidence. Data visualizations also enable audiences to explore the dataset and filter out information that is personally relevant. Engebretsen et al. (2018) also pointed out that employing exploratory data visualizations in storytelling can open the possibility for audiences to create many different stories or discover unseen stories.

中文译文：

数据可视化是新闻叙事的有力工具，能够发挥多种功能（Engebretsen et al., 2018; Lopezosa et al., 2023）。如果将数据可视化作为故事的核心元素，它就能够突出特定的新闻视角。当新闻报道中的主张需要事实支持时，数据可视化可以提供经验证据。数据可视化还使受众能够探索数据集，并筛选出与个人相关的信息。Engebretsen 等人（2018）也指出，在叙事中使用探索性数据可视化，可能使受众创造出多种不同的故事，或发现此前未被看见的故事。

[P013]

英文原文：

However, including data visualizations to improve storytelling is not simply throwing charts in a news story. Data visualizations should assist in the unfolding of a story, which keeps audiences engaged but does not disrupt the pace and structure. Telling a compelling news story requires authors to understand the narrative structure that is familiar to the audience (Rayfield, 1972). It was found that audiences will easily recognize and accept a story if it possesses a certain structure, which is characterized by some complexity. Completely explorative visualizations without guidance are not considered a data story as there are no annotations or explanations to help audiences capture the intended message (Lee et al., 2015). In addition to structure, the techniques used in data stories also play an important role in

conveying the intended story and affecting the interpretation of the audience. The rest of this chapter focuses on two major topics—the narrative visualization structure and visualization techniques—to discuss and conclude how previous scholars have discovered the optimal way to present data stories.

中文译文：

然而，通过数据可视化改进叙事，并不等于随意往新闻故事里添加图表。数据可视化应当辅助故事逐步展开，既保持受众的参与，又不打乱叙事节奏和结构。要讲述引人入胜的新闻故事，作者必须理解受众熟悉的叙事结构（Rayfield，1972）。研究发现，如果故事具有某种包含一定复杂性的结构，受众就更容易识别和接受它。完全没有引导的探索性可视化不能被视为数据故事，因为它缺少帮助受众把握预期信息的注释或解释（Lee et al.，2015）。除了结构，数据故事所使用的技术也在传递作者想讲述的故事以及影响受众的阐释方面发挥重要作用。本章余下部分将聚焦叙事可视化结构和可视化技术两个主要主题，讨论并总结既有学者如何发现呈现数据故事的最佳方式。

叙事可视化结构及其他结构

[P014]

英文原文：

Narrative visualization is defined as “the process by which the author achieves the goals of data visualization using narrative techniques” (Edmond & Bednarz, 2021). In journalism, narrative visualization is a more novel concept to academics than to practitioners: news media outlets have been integrating dynamic visualizations in news stories for a long time, whereas academic discussion on this topic remains relatively rare (Howard, 2014). In comparison, scholars from engineering, informatics, computer science, and HCI have been studying narrative visualization from the perspectives of narrative structures, tactics, approaches, techniques, and tools. Of all narrative visualization elements, the narrative structure serves as a framework on which authors build, order, and present the narrative to audiences. A story’s narrative structure can affect audiences’ perception, understanding, and memory of it (Yang et al., 2022). Visual design considerations also rely on the narrative structure of a story. Therefore, drawing on the literature from the abovementioned disciplines, I will discuss how journalists can enhance the efficacy of data storytelling by incorporating narrative visualization structures or other structures into data-informed stories.

中文译文：

叙事可视化被定义为“作者利用叙事技巧实现数据可视化目标的过程”（Edmond & Bednarz，2021）。在新闻业中，与从业者相比，叙事可视化对于学者而言是一个更新颖的概念：新闻媒体早已在报道中整合动态可视化，但有关这一主题的学术讨论仍然相对少见（Howard，2014）。相比之下，工程学、信息学、计算机科学及人机交互（HCI）领域的学者已经从叙事结构、策略、方法、技术和工具等角度研究叙事可视化。在叙事可视化的各个要素中，叙事结构是作者建构、组织并向受众呈现叙事的框架。故事的叙事结构能够影响受众对故事的感知、理解和记忆（Yang et al.，2022）。视觉设计方面的考量同样依赖故事的叙事结构。因此，我将借鉴上述学科的文献，讨论记者如何通过将叙事可视化结构或其他结构融入数据支持的故事，提高数据叙事的效果。

叙事可视化结构

[P015]

英文原文：

The structure of narrative visualization lies in the spectrum of author-driven and read-driven approaches (Segel & Heer, 2010). The author-driven approach, which is optimum for storytelling and efficient communication, follows a strictly linear structure through visualization and messaging and allows no

interactivity. In contrast, the reader-driven approach has a high degree of inter-activity and no strictly structured narrative path. In reader-driven stories, data visualizations function as a tool that gives audiences maximum freedom to explore and analyze the data and story (Weber et al., 2018). Most of the time, news stories use a combination of author-driven and reader-driven approaches, falling between the two sides of the spectrum. Journalists need to strike a balance between leaving room for interactivity and laying out a structured narrative path.

中文译文:

叙事可视化结构位于作者驱动（author-driven）与读者驱动（reader-driven）两种方法构成的连续谱上（Segel & Heer，2010）。作者驱动的方法最有利于讲述故事和高效沟通，它通过可视化和信息传递遵循严格的线性结构，不允许交互。相反，读者驱动的方法交互性很强，没有严格固定的叙事路径。在读者驱动的故事中，数据可视化作为一种工具，赋予受众探索和分析数据及故事的最大自由度（Weber et al.，2018）。大多数新闻故事将两种方法结合起来，处于连续谱的两端之间。记者需要在为交互保留空间和提供结构化叙事路径之间取得平衡。

[P016]

英文原文:

Hybrid combinations of two approaches are most seen in online news stories. In the hybrid structure, the main story is provided by the author, but audiences also have the freedom to explore the data in a certain part of the story (Weber et al., 2018). Segel and Heer (2010) identified three typical hybrid combinations, each of which has different permutations and emphasis and works for different types of stories.

中文译文:

这两种方法的混合组合在网络新闻故事中最为常见。在混合结构中，作者提供主线故事，但受众也可以自由探索故事中特定部分的数据（Weber et al.，2018）。Segel 与 Heer（2010）确定了三种典型混合形式，各有不同的变体和侧重点，适用于不同类型的故事。

[P017]

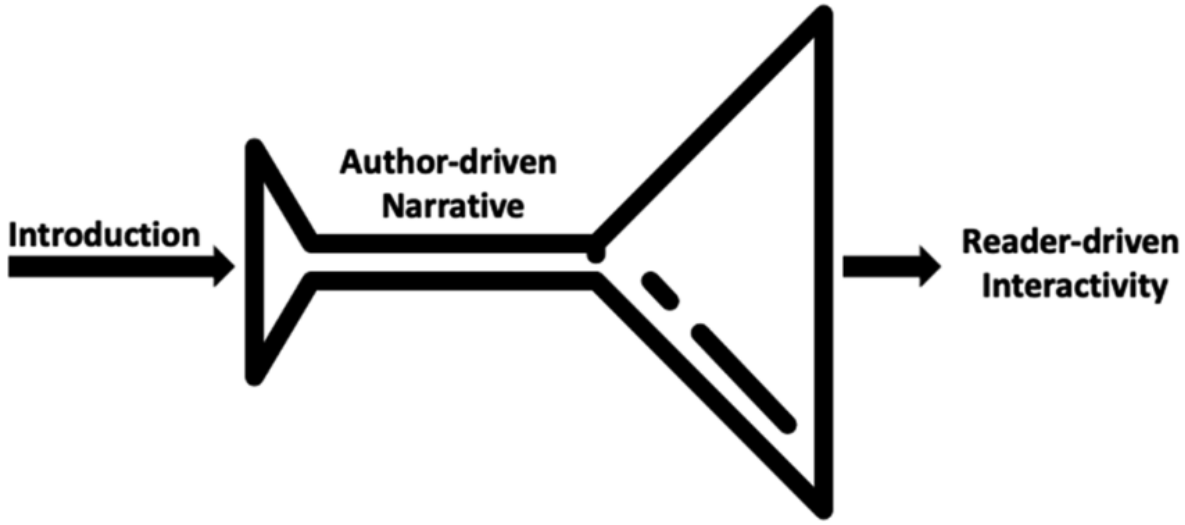
英文原文:

The Martini Glass structure begins with the author-driven approach, using questions, observations, or written articles to construct the author's predetermined sequences (Segel & Heer, 2010). Once the author-driven part is finished, the visualization opens up to the reader-driven part, inviting audiences to explore the data and interact with the display. Authors can use questions, observations, paragraphs, or articles in the author-driven part and use highlighting, filtering, or path-choosing in the reader-driven part. The Martini Glass structure is often encountered in interactive visualization news stories (see Fig. 18.1).

中文译文:

马提尼杯（Martini Glass）结构首先采用作者驱动的方法，以问题、观察或书面文章组织作者预先确定的顺序（Segel & Heer，2010）。作者驱动部分结束后，可视化进入读者驱动部分，邀请受众探索数据并与画面互动。作者可以在前一部分使用问题、观察、段落或文章，在后一部分采用突出显示、筛选或选择路径等方式。马提尼杯结构常见于交互式可视化新闻故事（见图18.1）。

[P018]



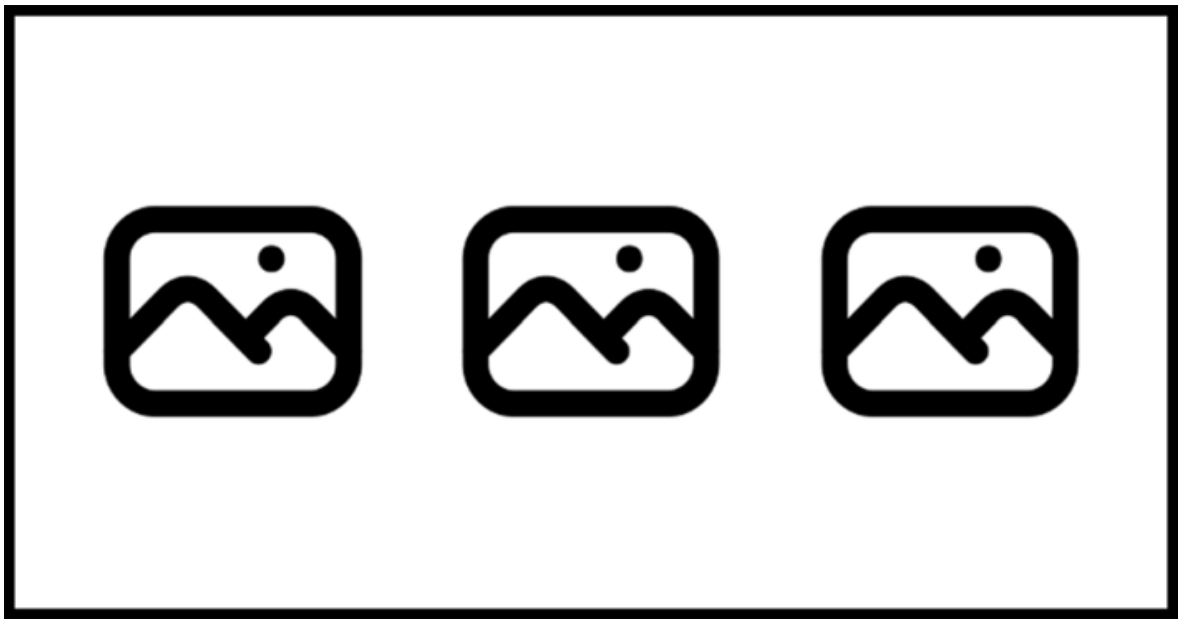
英文原文:

Fig. 18.1 Martini glass structure

中文译文:

图18.1 马提尼杯结构

[P019]



英文原文:

Fig. 18.2 Interactive slideshow structure

中文译文:

图18.2 交互式幻灯片结构

[P020]

英文原文:

The Interactive Slideshow structure follows a typical slideshow format but allows audiences to interact with some slides in the middle before moving forward to the next slide (Segel & Heer, 2010). Interestingly, in the interactive slideshow, each interactive slide is structured in Martini Glass style: beginning with an author-intended narrative and then following a visualization with which audiences

interact. The interactive slideshow is recommended to be used for multi-dimensional datasets or complex stories. Each slide can navigate audiences through the dimensions of a dataset and help them obtain a deeper understanding of each dimension through interaction. Audiences are given enough time to digest one slide's information before clicking on the next slide. For complex stories, the slideshow format provides audiences with a clear roadmap that enables audiences to distinguish different segments within a big story (see Fig. 18.2).

中文译文：

交互式幻灯片（Interactive Slideshow）结构遵循典型的幻灯片形式，但允许受众在进入下一张之前与中间的某些幻灯片互动（Segel & Heer，2010）。有趣的是，每张可交互的幻灯片都采用马提尼杯结构：从作者预设的叙事开始，再接续受众可以互动的可视化。交互式幻灯片适合多维数据集或复杂故事。每张幻灯片能够引导受众逐一认识数据集的不同维度，并通过互动加深对各维度的理解。受众有充分时间消化一页的信息，然后才点击下一页。对于复杂故事，幻灯片形式提供了清晰的路线图，使受众能够分辨大故事中的不同部分（见图18.2）。

[P021]

英文原文：

The Drill-Down Story structure is more reader-driven oriented (Segel & Heer, 2010). The Drill-Down Story visualization comprises a major theme and several particular sub-sections that let audiences click to reveal more additional details and backstories (see Fig. 18.3). Although the Drill-Down Story is more reader-driven inclined compared with the above two structures, it still needs authors to decide which sub-section on the visualization includes more additional details or backstories.

中文译文：

下钻式故事（Drill-Down Story）结构更偏向读者驱动（Segel & Heer，2010）。这种可视化包括一个主要主题和若干特定子部分，受众可以点击以揭示更多细节及背景故事（见图18.3）。虽然相较前两种结构，下钻式故事更加偏向读者驱动，作者仍须决定可视化中的哪些子部分需要包含额外细节或背景故事。

其他结构

积木堆叠结构 / The Stack of Blocks

[P022]

英文原文：

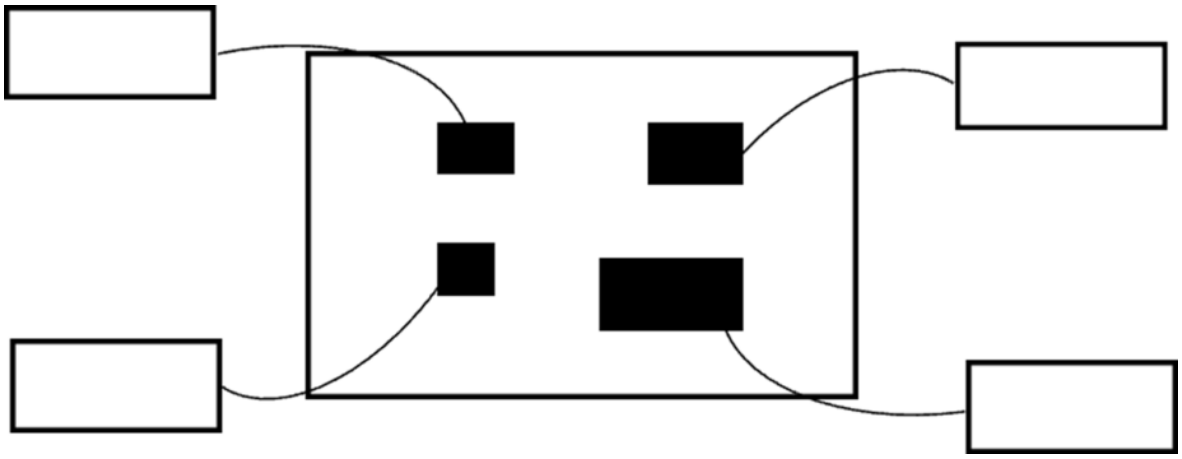
The Stack of Blocks is another storytelling structure that has been used by data journalists (see Fig. 18.4). The Stack of Blocks structure contains a beginning, middle, and end. The beginning provides the introduction to the story and predicts the middle, the middle is divided into sections by sub-topics in a logical order, and the end reiterates the main point and strengthens audiences' memories (Fry, 2009; Heravi, 2022). Different from the traditional narrative structure, the Stack of Blocks does not have a dramatic arc. In contrast to drawing attention to the character's journey and the chronology of the events, the Stack of Blocks put emphasis on the discussion of each sub-topic within the whole story in a logical order, which is advisable for data journalism. In Fig. 18.4, the gold coin between two parts indicates the place to insert data visualizations.

中文译文：

积木堆叠（Stack of Blocks）结构是数据新闻工作者使用的另一种叙事结构（见图18.4）。它包含开头、中间和结尾三个部分。开头介绍故事并预示中间部分；中间部分按照逻辑顺序划分为多个子主题；结尾重申主旨并强化受众的记忆（Fry，2009；Heravi，2022）。不同于传统叙事结构，积木堆叠没有戏剧性的情节

弧线。它不着重描绘人物经历和事件的时间顺序，而是强调依照逻辑顺序讨论整个故事中的各个子主题，这适用于数据新闻。图18.4中两个部分之间的金币标志表示插入数据可视化的位置。

[P023]



英文原文：

Fig. 18.3 Drill-down story structure

中文译文：

图18.3 下钻式故事结构

[P024]



英文原文：

Fig. 18.4 The stack of blocks structure

中文译文：

图18.4 积木堆叠结构

水塔结构 / The Water Tower Structure

[P025]

英文原文：

After analyzing 118 data-driven stories, Heravi (2022) proposed a new structure—Water Tower structure, which fits many data stories. In the introduction part, the water tower structure takes the Inverted Pyramid structure and includes the key facts. When moving to the middle part, it takes the Stack of Blocks structure's stem, grouping information by subjects and arranging them in a logical manner. The end also follows the Stack of Blocks structure, cementing the end into audiences' memories. Overall, the Water Tower structure is an improved version of the Stack of Blocks structure.

中文译文：

分析了118篇数据驱动报道后，Heravi（2022）提出了一种适用于许多数据故事的新结构——水塔（Water Tower）结构。引言部分采用倒金字塔结构，包含关键事实；进入中间部分后，则采用积木堆叠结构的主干，按主题对信息分组并以逻辑方式安排；结尾同样遵循积木堆叠结构，以加深受众对结尾的记忆。总体而言，水塔结构是积木堆叠结构的改进版本。

[P026]

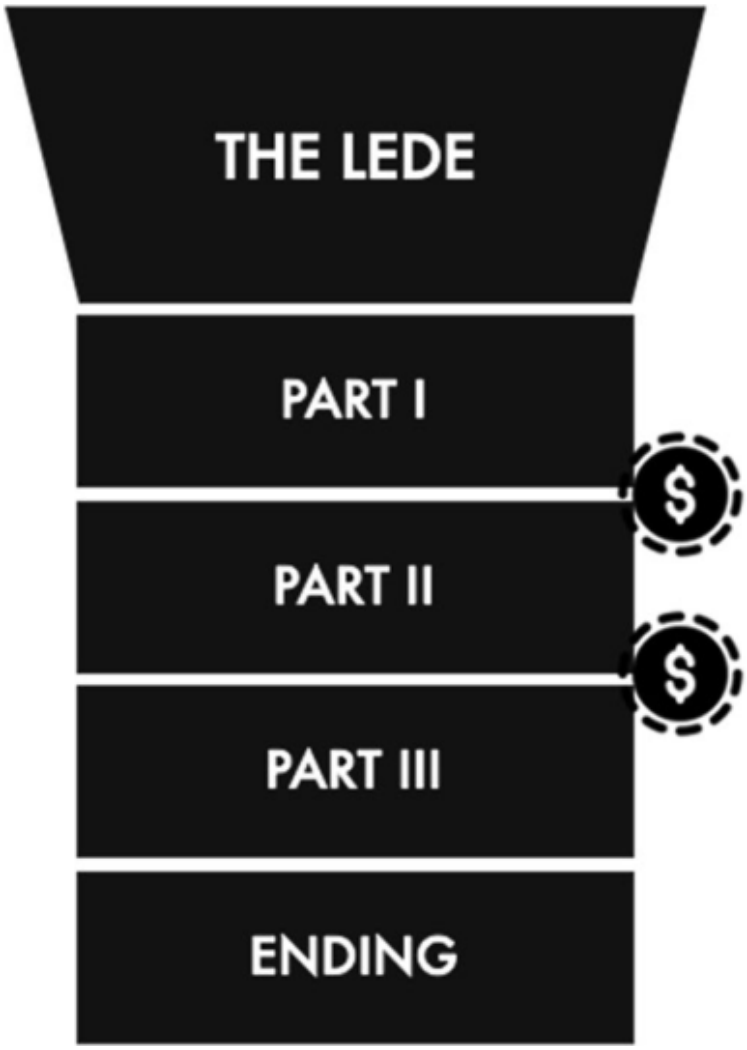
英文原文：

The Lede in the Water Tower structure follows the Inverted Pyramid structure and presents the most important facts as the hook to attract the audience's attention. As audiences move down to the middle of the story, the Water Tower structure takes them deeper into the 5Ws and 1H by discussing relevant sub-topics. The coin sign in Fig. 18.5 indicates the place where authors can insert data visualizations to keep audiences stay interested in the story.

中文译文：

水塔结构的导语沿用倒金字塔结构，将最重要的事实作为吸引受众注意的引子。当受众进入故事中部时，水塔结构通过讨论相关子主题，引导其更深入理解新闻的“五个W和一个H”（5Ws and 1H）。图18.5中的金币标志表示作者可以插入数据可视化的位置，以保持受众对故事的兴趣。

[P027]



英文原文：

Fig. 18.5 The water tower structure

中文译文：

图18.5 水塔结构

可视化技术

[P028]

英文原文:

Successful storytelling and engaging narrative visualization cannot be achieved without incorporating visualization techniques. Visualization techniques can influence audiences' experiences of interacting with an online article and understanding a story. Scrollytelling is a storytelling technique combining visuals and texts that change as audiences scroll through an online article. It is recommended for in-depth stories and can use a linear or elastic narrative structure. Annotations are important for guiding attention and conveying core messages. Visualization rhetoric is used to prioritize certain interpretations, and data visualization designers should create unambiguous designs. This section focuses on discussing these three major visualization techniques which have been recommended to use in data-driven stories.

中文译文:

成功的叙事和引人入胜的叙事可视化离不开可视化技术。可视化技术能够影响受众与网络文章互动以及理解故事的体验。滚动叙事是一种将视觉内容与文字结合起来的叙事技术，随着受众滚动网页，视觉内容和文字也发生变化。它适用于深度报道，可以采用线性或弹性叙事结构。注释对于引导注意力和传达核心信息十分重要。可视化修辞用来优先引导某些阐释方式，而数据可视化设计者应当创造没有歧义的设计。本节重点讨论推荐用于数据驱动故事的这三类主要可视化技术。

滚动叙事

[P029]

英文原文:

Scrollytelling, which combines the terms "storytelling" and "scrolling", refers to a storytelling technique to make visuals and texts change as audiences scroll through an online article (Seyser & Zeiller, 2018; Oesch et al., 2022). A typical scrollytelling article starts with full-screen photography and progresses to the next element as audiences scroll vertically or seldom horizontally (Seyser & Zeiller, 2018). Although a large number of visual elements are integrated into scrollytelling articles, text is still the backbone and provides major content. The scrollytelling technique is recommended for in-depth stories that cover complicated topics and plenty of details, such as investigative journalism or data-driven journalism.

中文译文:

滚动叙事（scrollytelling）结合了“讲故事”（storytelling）与“滚动”（scrolling）两词，指受众滚动在线文章时使视觉内容和文本相应变化的叙事技术（Seyser & Zeiller, 2018; Oesch et al., 2022）。典型的滚动叙事文章从全屏照片开始，随着受众垂直滚动（较少采用水平滚动）而进入下一个元素（Seyser & Zeiller, 2018）。虽然这类文章融合了大量视觉元素，但文字仍然是骨架，并承载主要内容。滚动叙事适用于涵盖复杂主题和丰富细节的深度故事，例如调查性新闻或数据驱动新闻。

[P030]

英文原文:

The scrollytelling technique suits either linear or elastic narrative structure; in other words, it can be used for either more author-driven or hybrid narrative visualization structure (Seyser & Zeiller, 2018). For an elastic narrative structure, the author still needs to have a predetermined order laid out for audiences but can branch off at specific stages of the story. These branches allow audiences to delve deeper into the story. Authors can also follow the structure of interactive slideshows or drill-down story: deconstruct a complex story into chapters, each of which follows an elastic structure. One caveat of using scrol-lytelling technique in a hybrid structure is that the author builds too many branches for audiences to find their way back to the main thread. Therefore, the author should ensure audiences can always go back to the core story no matter which branch they go to for more details.

中文译文:

滚动叙事技术既适合线性叙事结构，也适合弹性叙事结构；换言之，它既可用于较为作者驱动的叙事可视化，也可用于混合结构（Seyser & Zeiller，2018）。在弹性结构中，作者仍须为受众安排预定顺序，但可以在故事的特定阶段延伸出支线。这些支线使受众能够进一步深入故事。作者也可以遵循交互式幻灯片或下钻式故事的结构：将复杂故事拆分为若干章节，每章采用弹性结构。在混合结构中运用滚动叙事时，有一个需要警惕的问题，即作者可能设置过多支线，以致受众找不到返回主线的路径。因此，无论受众沿着哪条支线探索细节，作者都应确保他们始终能够回到核心故事。

[P031]

英文原文：

There are five scrollytelling techniques in total: animated transitions, graphic sequence, pan-and-zoom, show-and-play, and moviescroller.

中文译文：

滚动叙事技术共分为五种：动画转场（animated transitions）、图形序列（graphic sequence）、平移与缩放（pan-and-zoom）、展示与播放（show-and-play），以及电影式滚动（moviescroller）。

图形序列 / Graphic Sequence

[P032]

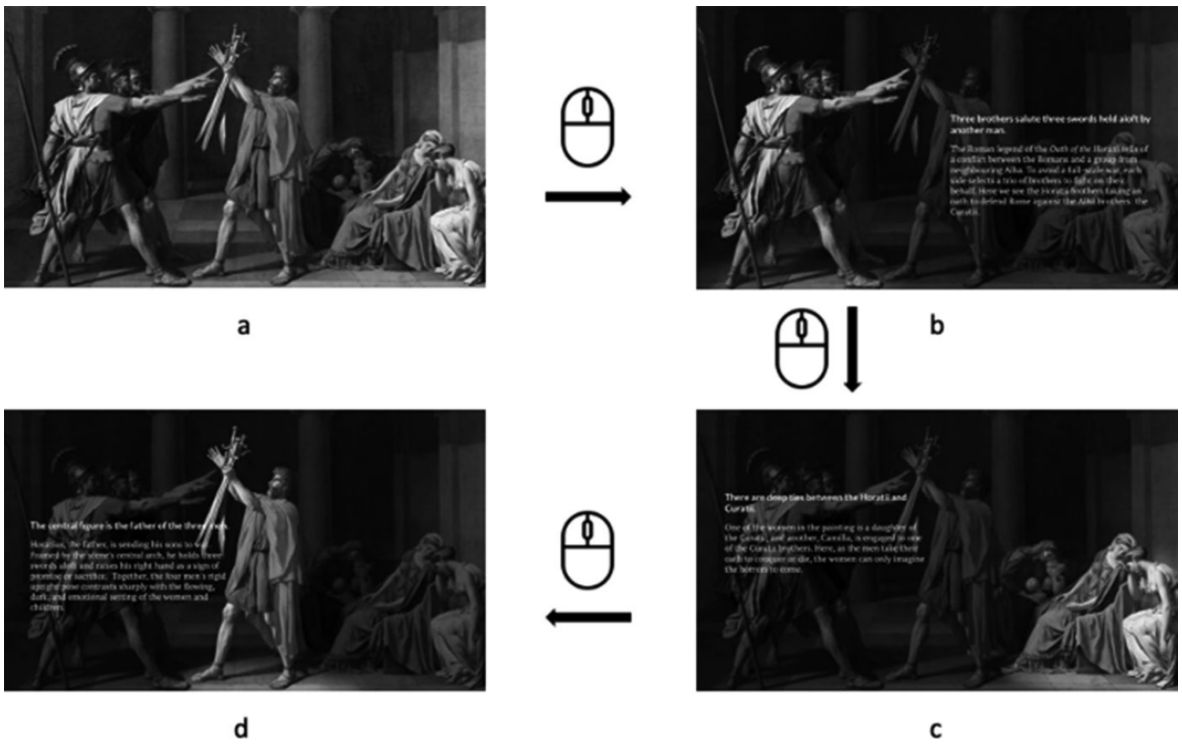
英文原文：

In a graphic sequence, images stay fixed in the background. As audiences scroll, text blocks move over the image, and the current image will transition into the next one with fading animation (Oesch et al., 2022) (see Fig. 18.6). This technique is recommended for highlighting certain parts of a visualization step by step. To keep the design consistent and clean, the image should be kept the same throughout the scroller.

中文译文：

在图形序列中，图像固定在背景上。受众滚动时，文字区块在图像上方移动，当前图像通过淡化动画过渡到下一幅图像（Oesch et al., 2022）（见图18.6）。这一技术适合逐步突出可视化的特定部分。为了保持设计的一致与整洁，整个滚动过程应使用相同的图像。

[P033]



英文原文：

Fig. 18.6 Graphic sequence scrollytelling technique. As users scroll, text blocks move over the image, and different parts of an image are highlighted

中文译文：

图18.6 图形序列滚动叙事技术。用户滚动时，文字区块在图像上方移动，图像的不同部分依次被突出显示。

动画转场 / Animated Transitions

[P034]

英文原文：

In animated transitions, images or charts stay fixed as text blocks scroll alongside (Oesch et al., 2022). As one text block scrolls up, the image or chart transitions into another with morphing animation. This technique works well for building up complex data visualizations in a stepwise fashion, adding each component stage by stage (see Fig. 18.7). It can also be used to show different aspects of a dataset by dismantling and regrouping symbols of a unit in a data visualization.

中文译文：

在动画转场中，图像或图表保持固定，文字区块在旁边滚动（Oesch et al., 2022）。当一个文字区块上移时，图像或图表通过变形动画转换为另一幅图像或图表。这一技术特别适合分步骤构建复杂的数据可视化，每一阶段添加一个组成部分（见图18.7）。它也可以通过拆分和重新组合可视化中一个单位的符号，展示数据集的不同方面。

平移与缩放 / Pan-and-Zoom

[P035]

英文原文：

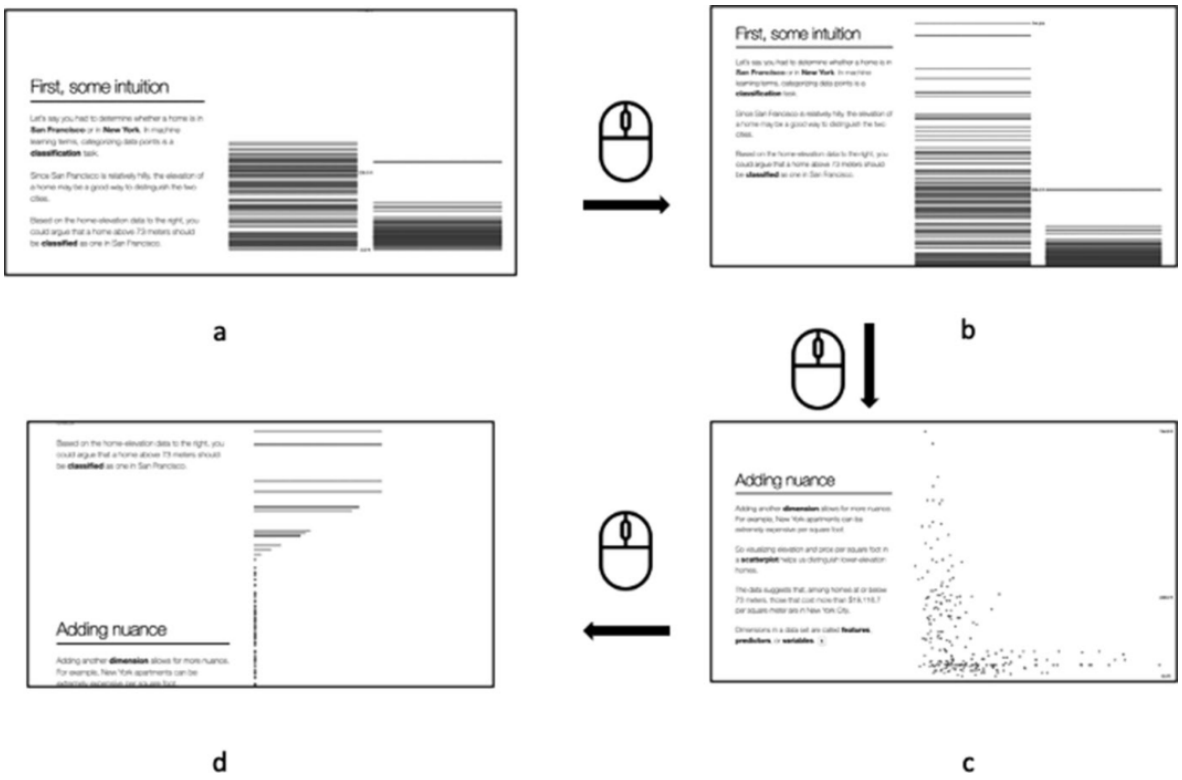
In pan-and-zoom transitions, one full-screen image with certain sections marked or highlighted for audiences (Oesch et al., 2022). As the audience scrolls, the image will zoom in to and enlarge one of the marked sections, displaying the necessary details of that section (see Fig. 18.8). Scrolling triggers the

image to zoom or pan. The speed of the zooming or panning movement depends on how fast audiences scroll. The pan-and-zoom technique is often used to help audiences navigate through maps or zoom in on different parts of an image.

中文译文：

在平移与缩放转场中，一幅全屏图像的某些区域被标记或突出显示，供受众关注（Oesch et a1., 2022）。受众滚动时，图像会放大其中一个标记区域，展示该区域所需的细节（见图18.8）。滚动触发图像缩放或平移；移动速度取决于受众滚动的快慢。这一技术常用于帮助受众浏览地图，或放大图像的不同部分。

[P036]



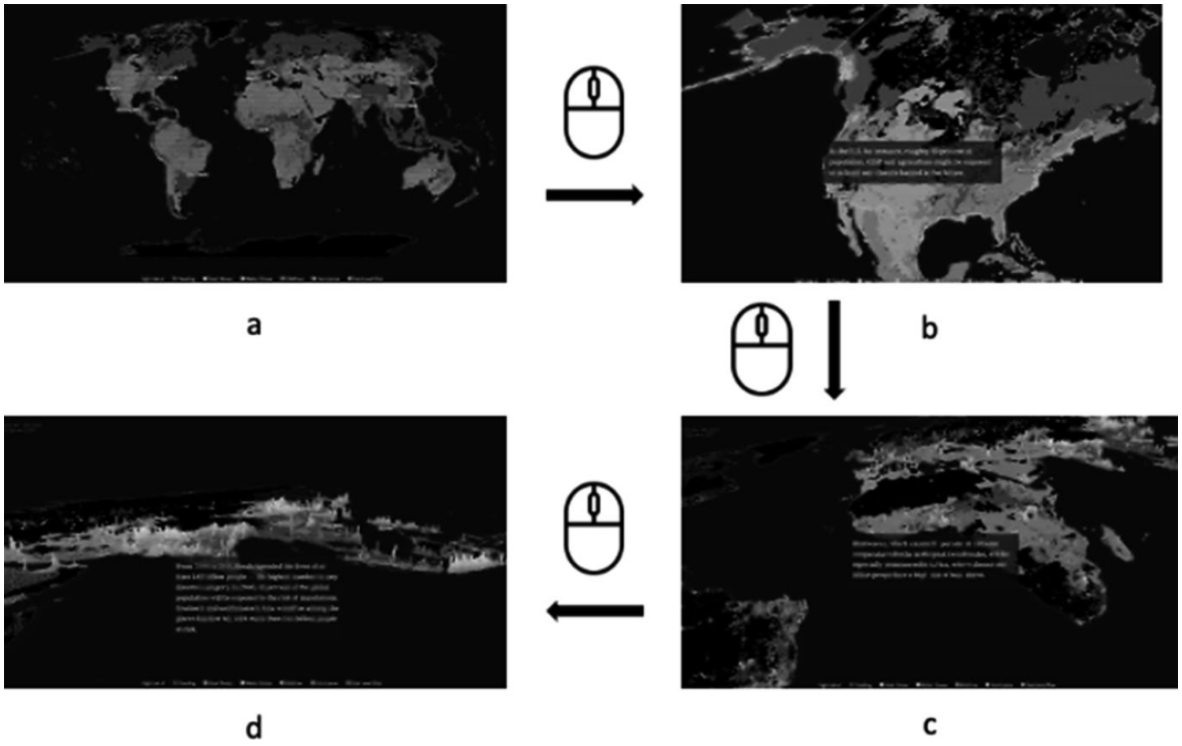
英文原文：

Fig. 18.7 Animated transitions scrollytelling technique. Text and image blocks stay in a fixed area. As text blocks scroll up, the image or visualization transitions into another with morphing animation

中文译文：

图18.7 动画转场滚动叙事技术。文字和图像区块保持在固定区域；文字区块向上滚动时，图像或可视化通过变形动画转换为另一幅。

[P037]



英文原文：

Fig. 18.8 Pan-and-Zoom scrollytelling technique. As users scroll, the map will zoom in to and enlarge one of the marked sections, displaying the necessary details of that section

中文译文：

图18.8 平移与缩放滚动叙事技术。用户滚动时，地图放大其中一个标记区域，并显示该区域所需的细节。

电影式滚动 / Moviescroller

[P038]

英文原文：

In moviescrollers, audiences scroll through a continuous sequence of images to create the moving effect (see Fig. 18.9). Moviescroller is often used to track shots in 3D models.

中文译文：

在电影式滚动中，受众通过滚动连续的图像序列制造运动效果（见图18.9）。电影式滚动通常用来追踪三维模型中的镜头。

展示与播放 / Show-and-Play

[P039]

英文原文：

Compared with the other four scrollytelling techniques, show-and-play is the least interactive technique. In the show-and-play, both textual and visual elements are placed in a predetermined order. The visual elements could be GIFs or videos, which will automatically play in a loop when audiences scroll to the part. Play-in-a-loop will stop when audiences scroll that visual element away.

中文译文：

与其他四种滚动叙事技术相比，展示与播放的交互性最低。在这种形式中，文字元素和视觉元素均按预先设定的顺序排列。视觉元素可以是GIF动态图或视频，当受众滚动到相应部分时，它们会自动循环播放；当该视觉元素滚出视野时，循环播放便会停止。

注释技术

[P040]

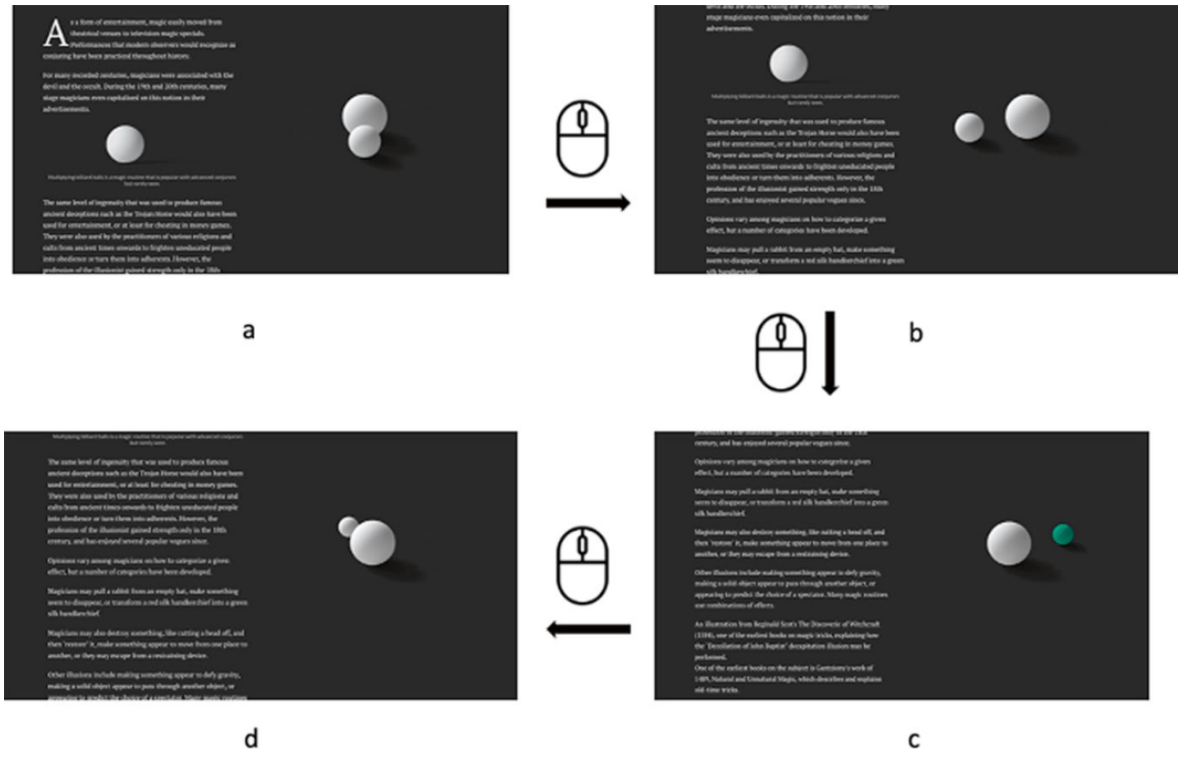
英文原文：

Annotations are often overlooked in data-driven storytelling, but they play a crucial role in conveying authors' core messages. By drawing attention to particular areas of a graph, annotations can help viewers better understand the data. Prior data visualization scholars also emphasized the importance of textual and graphical annotations in helping audiences capture the core messages in data-driven stories (Choe et al., 2015; Lee et al., 2015). Annotations are not only referring to textual commentaries that are added to a chart (Ren et al., 2017). An annotation can be any kind of superstructural elements that assists authors present insights and facilitate audiences' understanding and interpretation (Kirk, 2012; Marshall, 1997). Ren et al. (2017) summarized four forms of visual annotations that had been used in data-driven stories: text, shapes, highlights, and images. Different forms of annotations serve different purposes in data visualization.

中文译文：

注释在数据驱动叙事中经常被忽视，但在传达作者的核心信息方面起着关键作用。通过将注意力引向图表的特定区域，注释可以帮助读者更好地理解数据。既有数据可视化研究也强调，文字和图形注释对于帮助受众把握数据驱动故事的核心信息非常重要（Choe et a1.，2015；Lee et a1.，2015）。注释并不只是添加在图表上的文字评论（Ren et a1.，2017）；任何能够帮助作者呈现洞见、促进受众理解和解释的附加结构元素，都可以构成注释（Kirk，2012；Marsha11，1997）。Ren 等人（2017）总结了数据驱动故事中使用的四类视觉注释：文字、形状、突出显示和图像。不同的注释形式在数据可视化中有着不同的作用。

[P041]



英文原文：

Fig. 18.9 Moviescroller technique. Users scroll to trigger the moving and changing of the small white ball in the image

中文译文：

图18.9 电影式滚动技术。用户通过滚动触发图像中白色小球的移动与变化。

文字 / Text

[P042]

英文原文:

Data-driven text annotations can be used to indicate the values of chart elements (Ren et al., 2017). For example, in Fig. 18.10, the percentages of both darker and white points in each circle point are presented in data-driven text annotations in the dot pie chart. When only a limited number of elements are annotated in a visualization, the purpose is to draw audiences' attention to them before seeing other elements that do not have annotations. Non-data-driven text annotations are used to provide information context, orientation, or editorial comment (see Fig. 18.10). The typography in annotations could also help convey meaning or facilitate understanding, which will be further discussed in the linguistic rhetoric section.

中文译文:

数据驱动的文字注释可以用来标示图表元素的数值（Ren et a1., 2017）。例如，在图18.10的点状饼图中，每个圆点内深色和白色部分所占百分比都由数据驱动文字注释呈现。当可视化中只有少量元素带有注释时，其目的是让受众在注意其他未标注元素之前先关注这些元素。非数据驱动的文字注释则用于提供信息背景、方向指引或编辑评论（见图18.10）。注释的字体排版也可以帮助传递意义或促进理解，这将在语言修辞部分进一步讨论。

形状 / Shapes

[P043]

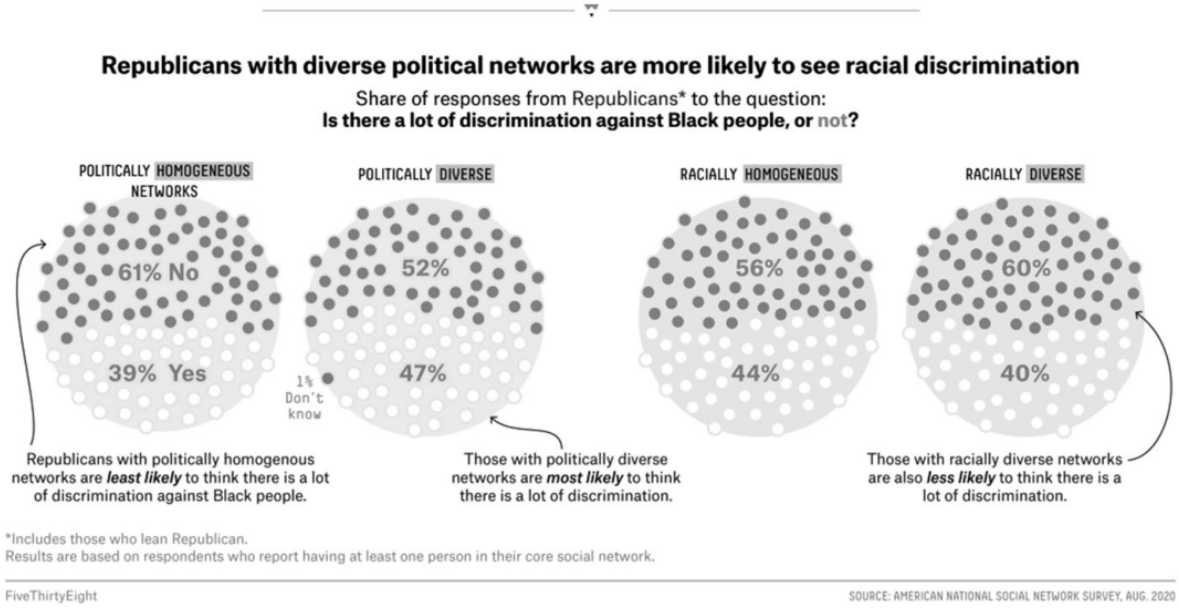
英文原文:

Shapes, such as rectangles, brackets, ellipses, arrows, lines, and speech bubbles, can also be used as annotations (Ren et al., 2017). Authors can use rectangles or ellipses to enclose chart elements, thereby prompting the viewer to recognize the significance of these elements or to contrast them with other outside elements. Arrows and symbols can be used to guide the audience's attention to a particular element. Shapes annotations could also be data-driven. For example, in a scatter plot, authors can add a trendline to indicate the underlying trend of two variables.

中文译文:

矩形、括号、椭圆、箭头、线条和对话气泡等形状，也可以充当注释（Ren et a1., 2017）。作者可以用矩形或椭圆圈出图表元素，促使观看者认识这些元素的重要性，或将它们与圈外其他元素比较。箭头和符号可以引导受众关注特定元素。形状注释也可以由数据驱动。例如，在散点图中，作者可以添加趋势线来表示两个变量之间的潜在趋势。

[P044]



英文原文：

Fig. 18.10 An example of using data-driven text to indicate data values and using non-data-driven text to provide information context context. Source: Koerth & Thomson-DeVeaux (2021)

中文译文：

图18.10 使用数据驱动文字表示数据数值，同时使用非数据驱动文字提供信息背景的示例。来源：Koerth & Thomson-DeVeaux（2021）。

突出显示 / Highlights

[P045]

英文原文：

Authors can also use highlights to emphasize the importance of the target. The highlight can be achieved by changing the element’s size, stroke, or color.

中文译文：

作者也可以通过突出显示强调目标元素的重要性。突出效果可以通过改变元素的大小、描边或颜色来实现。

图像 / Images

[P046]

英文原文：

Authors can add images and icons to the target to provide extra information and emphasize its importance. Images and icons can be distinguished by their size, opacity, or saturation, as well as their position in relation to the target. For example, in Fig. 18.11, authors used the images of senators and house representatives to indicate who reported a trade of stock or another financial asset by themselves.

中文译文：

作者可以为目标元素添加图片和图标，提供额外信息并强调其重要性。图片和图标可以通过大小、不透明度、饱和度以及相对于目标的位置来区分。例如，在图18.11中，作者使用参议员和众议员的照片，标明哪些人申报了自己进行的股票或其他金融资产交易。

可视化修辞技术

[P047]

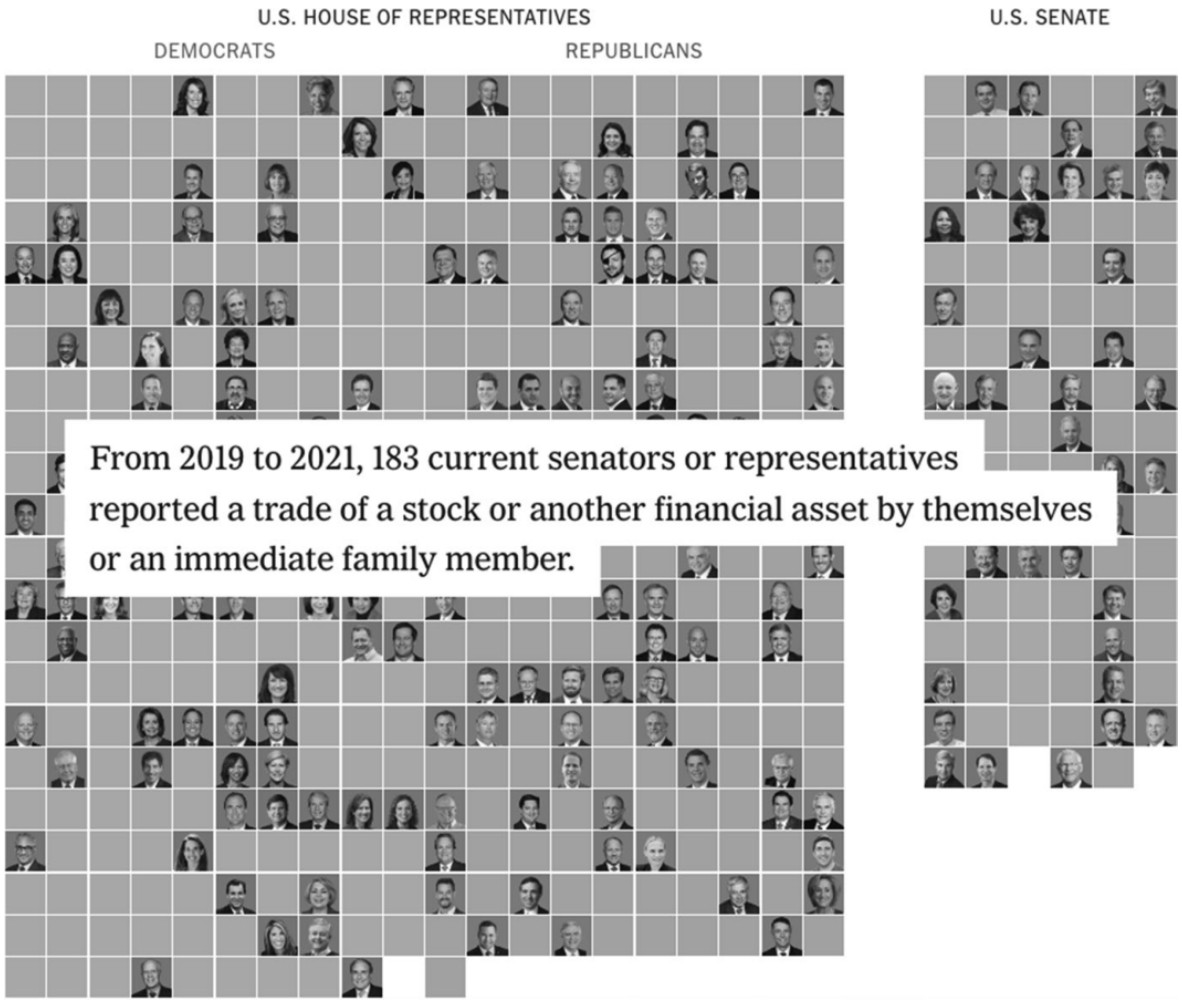
英文原文：

Given the interactive and explorative nature of narrative visualizations, which might lead to multiple interpretations on the audiences' side, visualization scholars have gone beyond studying the clarity and efficiency of information visualizations and started to look into how design strategies and rhetorical techniques can make certain interpretations more probable (Hullman & Diakopoulos, 2011). Visualization rhetoric refers to "the set of processes by which intended meanings are represented in the visualization via a designer's choices and then shaped by individual end-user characteristics, contextual factors involving societal or cultural codes, and the end-user's interaction" (Hullman & Diakopoulos, 2011, p. 2232). Prior scholars found that design elements that did not directly communicate the designers' intended meaning could bias audiences' interpretations of the designers' original meaning (Norman, 1999). This is because implicit meaning encoded in visual design may interact with audiences' knowledge and understanding of conventions, which subsequently influence their interpretation. Due to the bias caused by visualization design, data visualization designers should create unequivocal designs in which the intended meaning is transparent and graphs are absolutely unambiguous (Kosslyn, 2006). To achieve this goal, data visualization designers should be equipped with knowledge of how to use certain design elements strategically to prioritize certain interpretations and skills of designing narrative visualizations to communicate layered meanings without ambiguity.

中文译文：

鉴于叙事可视化具有交互性与探索性，可能使受众产生多种不同的解释，可视化研究者已不再只考察信息可视化的清晰度与效率，而是开始研究设计策略和修辞技术如何使特定解释更有可能出现（Hullman & Diakopoulos，2011）。可视化修辞（visualization rhetoric）是指这样一组过程："预期意义通过设计者的选择在可视化中得到表达，随后受到最终用户的个人特征、涉及社会或文化惯例的情境因素及其交互行为的塑造"（Hullman & Diakopoulos，2011，第2232页）。既有研究发现，未能直接传达设计者预期意义的设计元素，可能使受众对设计者原意的理解产生偏差（Norman，1999）。这是因为视觉设计中编码的隐含意义可能与受众对惯例的知识和理解相互作用，进而影响其解释。鉴于可视化设计可能引发偏差，数据可视化设计者应创造意图透明、图表绝无歧义的明确设计（Kosslyn，2006）。为实现这一目标，设计者需要掌握策略性运用特定设计元素以优先引导某些解释的知识，以及设计叙事可视化以清楚传递多层意义的技能。

[P048]



英文原文：

Fig. 18.11 An example of using images to provide extra information. Source: Kelly, Playford, Parlapiano, & Uz (2022)

中文译文：

图18.11 使用图片提供额外信息的示例。来源：Kelly、Playford、Parlapiano与Uz（2022）。

映射修辞

[P049]

英文原文：

Mapping rhetoric refers to the manipulation of information presentation through data-visual transmission functions and the constraints that decide how information is converted to visual elements (Hullman & Diakopoulos, 2011). Obscuring can occur by introducing noise into representations at the perception level, such as unnecessary size transformations. Visual elements that are too small for judgment, oversizing, or obscure the actual position of the value on an axis will all disrupt the audiences' interpretation of the visualization. Besides obscuring, Hullman and Diakopoulos (2011) also introduced three mapping rhetoric techniques that can be used in facilitating audiences' decoding and understanding of visualizations.

中文译文：

映射修辞（mapping rhetoric）指通过数据—视觉转换功能，以及决定如何将信息转换为视觉元素的约束条件，来操控信息的呈现方式（Hullman & Diakopoulos，2011）。在感知层面向表征加入噪声，例如不必要

的尺寸变化，可能造成信息遮蔽。视觉元素过小而难以判断、尺寸过大，或掩盖数值在坐标轴上的真实位置，都会干扰受众对可视化的解释。除遮蔽之外，Hullman与Diakopoulos（2011）还介绍了三种映射修辞技术，可以帮助受众解码和理解可视化。

视觉隐喻 / Visual Metaphor

[P050]

英文原文：

In linguistics, metaphors are used to help audiences understand a primary and abstract concept by associating it with a secondary concept that is familiar and tangible to audiences (Borgo et al., 2012). Visual metaphor is a technique where a primary and abstract concept is conveyed by making good use of visual elements that are familiar to audiences. For example, computers use an icon of a paper folder, which is similar to the file folder in the real world, to teach users how to use it to store and organize files. Besides facilitating understanding, visual metaphors can also draw audiences' attention and strengthen the memory of presented information.

中文译文：

在语言学中，隐喻通过将主要的抽象概念与受众熟悉、具体可感的次要概念联系起来，帮助他们理解前者（Borgo et al., 2012）。视觉隐喻（visual metaphor）则利用受众熟悉的视觉元素来传达主要的抽象概念。例如，计算机使用与现实世界文件夹相似的纸质文件夹图标，教用户如何存储和整理文件。视觉隐喻除了促进理解，也能够吸引受众注意，并强化对呈现信息的记忆。

[P051]

英文原文：

In data visualization stories, authors can also use the visual metaphor technique to convey implicit and abstract meanings and strengthen the memory of visualized content. More concretely, authors can use spatial mappings, typo-graphic mappings, and color mappings (Hullman, 2020). For example, in election-themed stories, authors can use blue and red font colors to represent the republican and democratic parties.

中文译文：

在数据可视化故事中，作者还可以使用视觉隐喻传达隐含和抽象的意义，并增强对可视化内容的记忆。更具体地说，作者可使用空间映射、字体映射和颜色映射（Hullman，2020）。例如，在选举主题的报道中，作者可以使用蓝色和红色字体分别代表共和党和民主党。

分类 / Classification

[P052]

英文原文：

Classification can be achieved through the categorization of data based on size, position, or color. Furthermore, the utilization of typographic manipulation of font sizes, styles, and equations of significance, presented in a legend-like format, can be used to emphasize certain values and thus classify the information within a visualization. This type of classification can be used to demonstrate clusters of priority or importance by making certain points stand out. Additionally, it can also help to organize data into easily understandable sections. For example, in Fig. 18.12, different shades and enclosed circles are employed to categorize 50 states that were required to get a gubernatorial recall. In this way, classification can be used to highlight patterns and trends contained within a dataset that may not be visible at first glance.

中文译文：

分类（classification）可以通过根据大小、位置或颜色对数据进行归类实现。此外，还可以采用图例式的字体处理方式，调整字号、样式及重要性表达，突出特定数值，从而对可视化中的信息进行分类。这种分类通过使某些数据点格外醒目来展示优先级或重要性的聚集，也有助于将数据组织为容易理解的部分。例如，图18.12使用不同深浅的颜色和封闭圆圈，对涉及州长罢免要求的50个州进行分类。由此，分类可以突出数据集中那些乍看之下未必可见的模式和趋势。

冗余 / Redundancy

[P053]

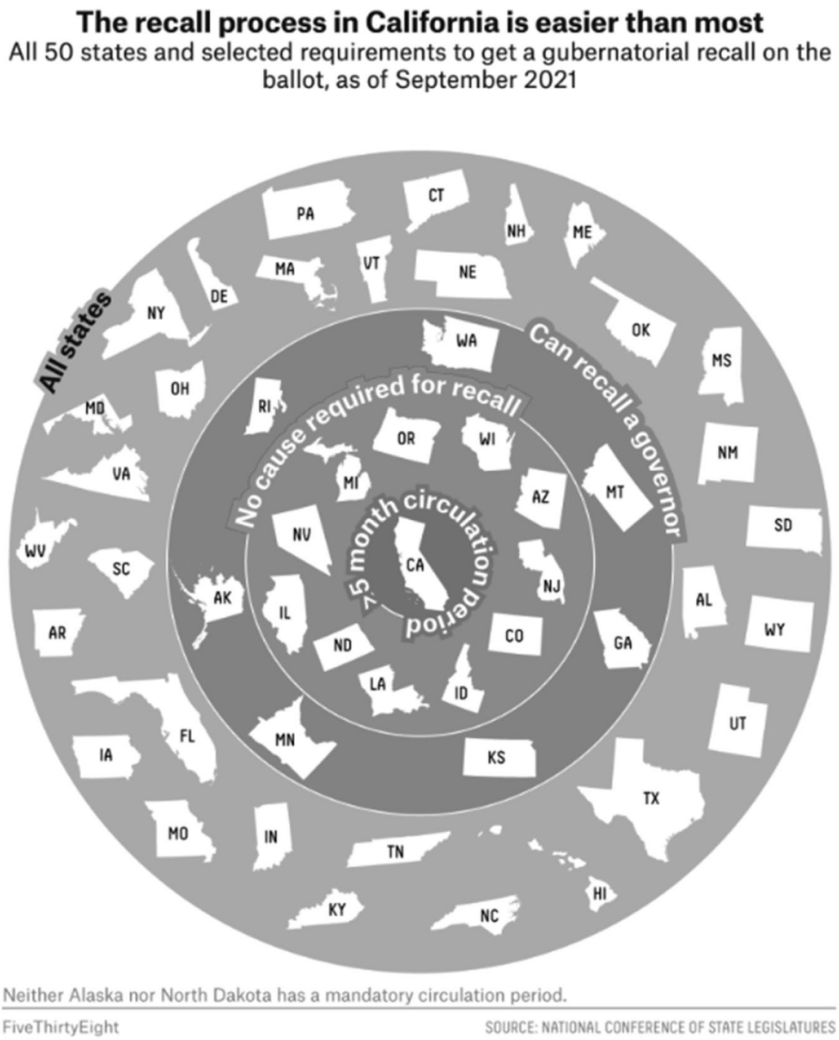
英文原文：

It has been observed that employing redundancy techniques can be beneficial in emphasizing homogenous values or visual marks (Hullman & Diakopoulos, 2011). This method can involve the utilization of identical labels or the disag-gregation of values with minimal variance. Additionally, employing the redundancy technique can highlight certain information in a visualization.

中文译文：

研究发现，使用冗余（redundancy）技术有助于强调相同类型的数值或视觉标记（Hullman & Diakopoulos，2011）。这一方法可以采用相同的标签，或将差异极小的数值拆开呈现。此外，冗余技术也能突出可视化中的特定信息。

[P054]



英文原文：

Fig. 18.12 An example of using color, shades, and different size of circles to categorize data. Source: Mithani & Sweedler (2021)

中文译文：

图18.12 使用颜色、深浅和大小不同的圆圈对数据进行分类的示例。来源：Mithani & Sweedler（2021）。

语言修辞

[P055]

英文原文：

The effects of texts in data visualizations have been underrated. However, text formatting and language usage in data visualizations could also shape authors' intended meaning. Titles, subtitles, labels, and annotations on a chart could also influence audiences' interpretation of the visualization. Hullman and Diakopoulos (2011) proposed four linguistic rhetoric techniques that have been seen in data-informed news stories.

中文译文：

文字在数据可视化中的作用一直被低估。然而，数据可视化中的文字排版与语言运用也能够塑造作者想表达的意义。图表中的标题、副标题、标签和注释，都可能影响受众对可视化的解释。Hullman与Diakopoulos（2011）提出了四种已见于数据支持的新闻故事中的语言修辞技术。

字体排版 / Typography

[P056]

英文原文:

Texts are integral to data visualizations and function at multiple editorial levels (Moroianu, 2021). Besides ensuring the readability and legibility of texts, authors can also create visual hierarchy through typography. The choice of typeface, text color, size, shape, and boldness can all serve to convey the author's meaning. For example, by choosing the size of the text, authors decide what audiences should see first, second, and last (Muth, 2022). Text formatting can be used to signal importance, structure, or performance in data visu-alizations. For example, choosing different typefaces or sizes for titles, headings, subheadings, and body can help audiences quickly pick up the structure of the chart and eliminate confusion.

中文译文:

文字是数据可视化的重要组成部分，在多个编辑层次发挥作用（Moroianu，2021）。除了确保文字易于阅读和辨认，作者也可以通过字体排版（typography）建立视觉层级。字体、文字颜色、字号、字形和粗细的选择，都可以传达作者的意图。例如，通过选择字号，作者可以决定受众首先、其次和最后看到什么（Muth，2022）。文字排版可用于表示信息的重要性、结构或表现形式。比如，为标题、章节标题、小标题及正文使用不同字体或字号，能够帮助受众迅速把握图表结构，减少困惑。

反讽 / Irony

[P057]

英文原文:

As a linguistic and literary device, irony is used where there is a contrast between literal meaning and implied meaning. Using irony in visualization titles can effectively attract audiences' attention. Using irony to phrase a rhetorical question implies its inverse and guides audiences to arrive at an obvious interpretation. For example, in Fig. 18.10, the subtitle includes a rhetorical question "Is there a lot of discrimination against Black people, or not?", intriguing audiences to look deeper into the below dot pie charts and draw their conclusions.

中文译文:

反讽（irony）是一种语言和文学手法，用于字面意义与隐含意义形成反差的情形。在可视化标题中运用反讽，可以有效吸引受众注意。以反讽方式提出修辞性问题，暗示与字面相反的含义，并引导受众得出一个明显的解释。例如，在图18.10中，副标题包含一个反问："针对黑人的歧视究竟多不多？"这会激发受众进一步观察下方的点状饼图，并得出自己的结论。

相似性 / Similarity

[P058]

英文原文:

Similarity techniques are applied in comparison situations where two entities are assumably similar. Authors can use analogy to explain the unclear meaning of two entities by comparing with two well-known entities (Hullman & Diakopoulos, 2011). Authors can also use the metaphoric statement to directly assert that one entity is the other, as in the data visualization (see Fig. 18.14) titled "Least Vaccinated U.S. Counties Have Something in Common: Trump Voters". Another similarity technique is Parallelism, displaying two similar statements or visual features to show they are equally important. For example, still in Fig. 18.13, the juxtaposition of two similar scatterplots is framed around the relationship between people's political stances and their vaccination hesitancy.

中文译文:

相似性（similarity）技术应用于两个实体被假定具有相似性的比较情境。作者可以通过将两个意义不明确的实体与两个广为人知的实体进行类比，解释前者（Hullman & Diakopoulos，2011）。作者也可以用隐喻性陈述直接宣称一个实体就是另一个实体，例如数据可视化（见图18.14）的标题“美国疫苗接种率最低的县有一个共同点：特朗普选民”。另一种相似性技术是平行法（parallelism），将两个相似的陈述或视觉特征并列展示，以表明它们同等重要。例如，同样在图18.13中，并置两幅相似的散点图，以呈现人们的政治立场与疫苗接种犹豫之间的关系。

个体化 / Individualization

[P059]

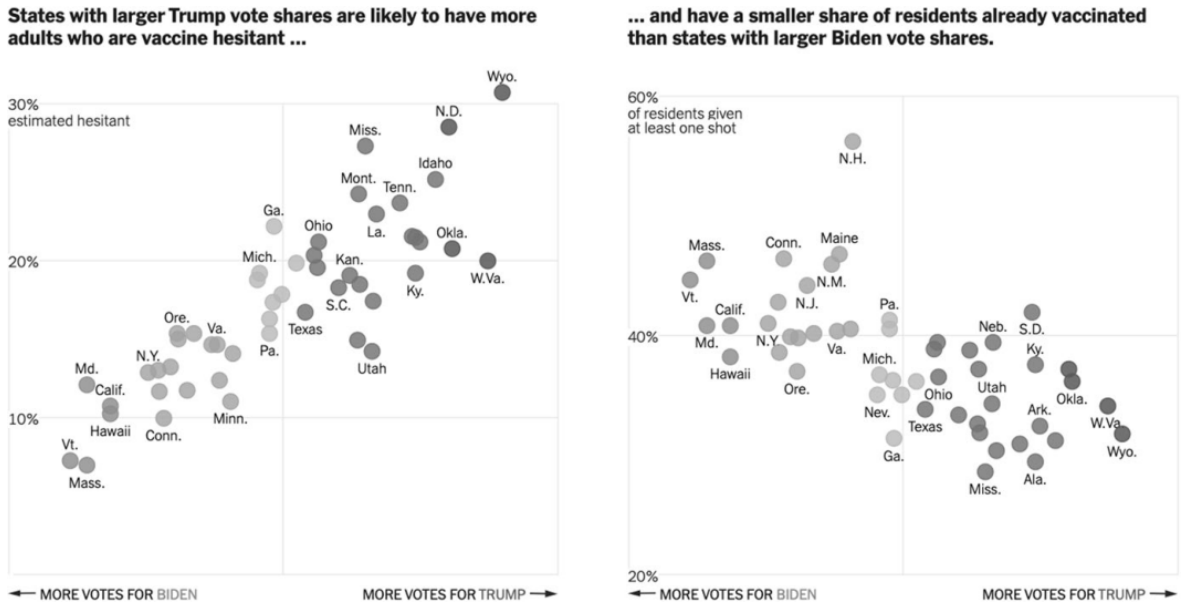
英文原文:

The individualization techniques aim to directly appeal to the audience as an individual, which can increase the interest and the sense of participation of the audience (Hullman & Diakopoulos, 2011). One subtle means to achieve that is to provide the audience with exploratory functions in data visualization, such as sorting and filtering methods. For example, in Fig. 18.14, the data visualization provides a filter method that allows users to select the state they are inter-ested in to explore the situation of administered doses in each state. Another popular means of individualization is to provide an interactive format of question and answer, where the story moves forward based on the audiences' answers to the questions.

中文译文:

个体化（individualization）技术旨在直接将受众作为个体加以吸引，从而提高其兴趣和参与感（Hullman & Diakopoulos，2011）。一种较为含蓄的方式，是在数据可视化中提供排序、筛选等探索功能。例如，图18.14提供筛选功能，允许用户选择自己感兴趣的州，查看各州已经接种的疫苗剂次数量。另一种常见的个体化方法，是提供问答式互动形式，故事根据受众对问题的回答继续发展。

[P060]



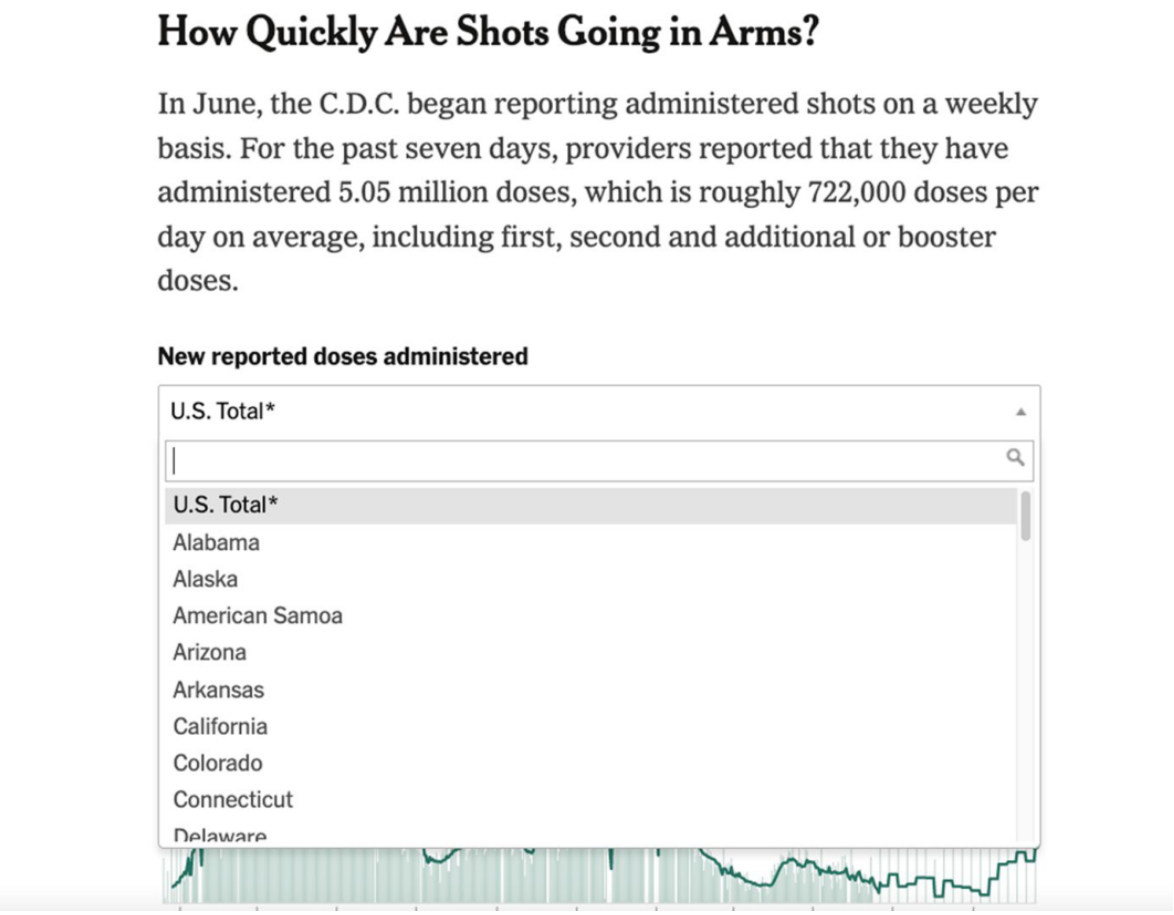
英文原文：

Fig. 18.13 Least Vaccinated U.S. Counties Have Something in Common: Trump Voters. An example of juxtaposing two similar scatterplots to represent the relationship between people's political stances and their vaccination hesitancy. Source: Ivory, Leatherby & Gebeloff (2021)

中文译文：

图18.13 “美国疫苗接种率最低的县有一个共同点：特朗普选民”。并置两个相似的散点图以呈现人们政治立场与疫苗接种犹豫之间关系的示例。来源：Ivory、Leatherby与Gebeloff（2021）。

[P061]



英文原文：

Fig. 18.14 An example of providing a filter method that allows users to select the state they are interested in to explore the situation of administered doses in each state. Source: The New York Times. See how vaccinations are going in your county and state

中文译文：

图18.14 提供筛选功能，让用户选择感兴趣的州，查看该州疫苗接种剂次数量情况的示例。来源：《纽约时报》“查看你所在县和州的疫苗接种进度”。

结论

[P062]

英文原文：

This chapter overviews previous literature on data storytelling, narrative visualization, and visualization techniques to offer a clear methodological view of how data and digital journalists can incorporate these strategies and techniques to improve data-driven storytelling. Nevertheless, the suggestions mentioned in this chapter do not intend to be either universal or prescriptive. The choice of narrative visualization structures, scrollytelling techniques, or linguistic techniques depends on many factors including the type of news story being prepared, the publishing platform, and the topic. For example, if a journalist is writing a breaking news story reporting the mortality rate in an earthquake, the Inverted Pyramid is better than any type of structure abovementioned. If the story is the result of a long-term, thorough investigation of a particular topic and integrates multiple datasets to conduct extensive analysis and investigation, then structures such as block stacks, Martini glass, and new water tower structures may be more appropriate. Therefore, not any data-driven story should go with the water tower or Martini structure, journalists need

to consider multiple factors that might influence how a story unfolds when selecting the most appropriate structure.

中文译文：

本章回顾了数据叙事、叙事可视化和可视化技术方面的既有文献，旨在清楚呈现一种方法论视角，说明数据新闻与数字新闻工作者如何整合这些策略和技术以改进数据驱动叙事。不过，本章提出的建议既非普遍适用，也不意在成为必须遵守的规定。选择何种叙事可视化结构、滚动叙事技术或语言技术，取决于所准备的新闻故事类型、发布平台和主题等诸多因素。例如，如果记者撰写的是一篇有关地震死亡率的突发新闻报道，倒金字塔结构会比上述任何结构更合适。如果故事是针对特定主题开展长期深入调查的成果，且整合多个数据集进行广泛分析和调查，那么积木堆叠、马提尼杯和新水塔结构可能更为合适。因此，并非任何数据驱动的故事都应采用水塔或马提尼杯结构；记者在选择最适合的结构时，需要考虑可能影响故事展开方式的多种因素。

[P063]

英文原文：

Similarly, when applying scrollytelling techniques, journalists need to consider the time, cost, and labor before creating scrollers for a long-form story. Although the scrollytelling technique has gained popularity in newsrooms, it is labor-intensive and expensive.

中文译文：

同样，在应用滚动叙事技术时，记者也需要在为长篇报道制作滚动效果之前考虑时间、成本和人力投入。尽管滚动叙事技术在新闻编辑室中日益流行，它仍然是一种劳动密集且成本高昂的技术。

[P064]

英文原文：

How to use emerging technology to improve data-driven storytelling and engage audiences is a broad topic involving journalists, design, disseminating platforms, and audiences. This chapter concentrates only on the niche of visualization design and technique. Given that audiences have more agency in exploring the narrative of a news story, future studies can discuss more on how audiences engage in different structured stories and interact with different medium interfaces. As technology advances, data-driven storytelling continues to evolve, and journalists or designers continue developing new approaches to support stories with data visualization and better inform the public. In the context of future journalism, scholars need to continue discussing how the latest technology will transform the formats of journalism storytelling and professional practices, change the role of audiences in news consumption and production, and the relationship between journalists and audiences.

中文译文：

如何利用新兴技术改进数据驱动叙事并吸引受众，是一个涉及记者、设计、传播平台和受众的广泛议题。本章只关注可视化设计和技术这一较为狭窄的领域。鉴于受众在探索新闻故事叙事时拥有更大的自主权，未来研究可以进一步讨论受众如何参与不同结构的故事，以及如何与不同媒介界面互动。随着技术进步，数据驱动叙事将继续演变，记者或设计者也会不断开发新方法，利用数据可视化支持叙事、更好地向公众传递信息。在未来新闻业的背景下，学者需要持续讨论最新技术将如何改变新闻叙事形式和专业实践，如何改变受众在新闻消费与生产中的角色，以及记者与受众之间的关系。

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本节保留原始英文参考文献列表的排版及可检索文字，未逐条翻译。

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